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The GLOBOCAN 2022 cancer estimates: Data sources, methods, and a snapshot of the cancer burden worldwide

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Abstract

The data sources and methods used to develop global cancer incidence and mortality statistics—the GLOBOCAN estimates—for the year 2022 are documented in this article, alongside a brief overview of the global cancer burden. The estimates, made available in 185 countries or territories worldwide for 36 cancer sites by sex and age, are based on the best available local data sources, namely population-based cancer registries (for incidence) and national vital statistics (for mortality). In males, lung cancer was the most commonly diagnosed cancer worldwide in 2022 (1.57 million new cases [95% UI: 1.56–1.58]), followed by prostate cancer (1.47 million [1.46–1.48]). With 2.30 million (2.28–2.30) new cases estimated in 2022, breast cancer was the most diagnosed cancer in females, followed by lung cancer (0.91 million [0.90–0.91 million]) and cervical cancer (0.66 million [0.66–0.67]). The most common causes of cancer death in males and females were lung cancer (1.23 million [1.22–1.24]) and breast cancer (0.67 million [0.66–0.67]), respectively.

KEYWORDS

cancer, global estimates, GLOBOCAN, incidence, mortality

What's New?

This report details the data sources and methods used to build IARC's GLOBOCAN database of cancer incidence and mortality estimates in 185 countries or territories for the year 2022. The core methods remain unchanged from previous iterations, and accord with the principles of transparency and reproducibility at the national level. Uncertainty intervals (UIs) for the estimates are provided to account for major sources of error over and above random variation, namely the timeliness, representativeness and quality of the underlying data sources.

1 | INTRODUCTION

Global cancer statistics are disseminated by the International Agency for Research on Cancer (IARC) as per its mandate as the specialized cancer agency of the World Health Organization (WHO).[1–4] These statistics serve to inform our understanding the global and regional burden of cancer and are a critical resource in developing and prioritising effective cancer control policies. As with previous reports,[5–7] we document here the data sources and methods used to estimate global cancer statistics for the year 2022. A comprehensive analysis of the cancer incidence and mortality burden by world region, with a focus on the 10 most common cancers globally, is available elsewhere.[3]

The methods of estimation for GLOBOCAN 2022 are conducted using the most valid data available on cancer incidence and mortality

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at national and subnational levels. The primary unit for these estimates is at the country level, and estimates are aggregated to world regions and according to predefined levels of human development. Uncertainty intervals have been estimated based on three indicators, namely the quality, timeliness and coverage of the source information within each country. The estimates are made available at IARC's Global Cancer Observatory (GCO, https://gco.iarc.who.int/en).

2 | METHODS

2.1 | Data sources

A primary source for IARC's global cancer statistics is the recorded incidence data collected by population-based cancer registries (PBCR) worldwide, including the high-quality data compiled in the quinquennial review, *Cancer Incidence in Five Continents* (CI5), now in its 12th Volume and covering cancer diagnoses 2013–2017. While robust recorded cancer incidence data is not available in all countries of the world, there have been steady increases in data availability in some regions and countries, such as in Sub-Saharan Africa via the African Cancer Registry Network (http://afcrn.org), and in China where the number of cancer registries has exponentially increased and now almost cover the entire country. National mortality data by cause from WHO are also a major source of information on cancer deaths, where available, and of sufficient completeness and coverage.[8]

As with previous estimates, the definitions of countries and world regions follow the UN country classification,[9] except for Cyprus, which is included in Southern Europe rather than Western Asia. National population estimates for 2022 were obtained from the UNDP website World Population Prospects.[9] Details of the data sources and methods by country can be found in *Annex A* (and at https://gco.iarc.who.int/today/en/data-sources-methods).

2.2 | Methods of estimation

The counts of new cancer cases, cancer deaths and population a for incidence and mortality rates by sex and 18 age groups (0–4, 5–9, 10–14, ..., 75–79, 80–84, and 85+) were estimated for the 185 countries and territories with populations of over 150,000 inhabitants in 2022. Incidence and mortality were estimated for all sites combined, and for all sites combined excluding non-melanoma skin cancers, as well as for 36 individual cancer sites, classified according to the International Classification of Diseases 10th Revision (ICD-10). The full list of cancer sites is provided in *Annex B*. The methods used for estimating incidence and mortality, along with their respective uncertainty intervals, are documented below. Further details on the modelling

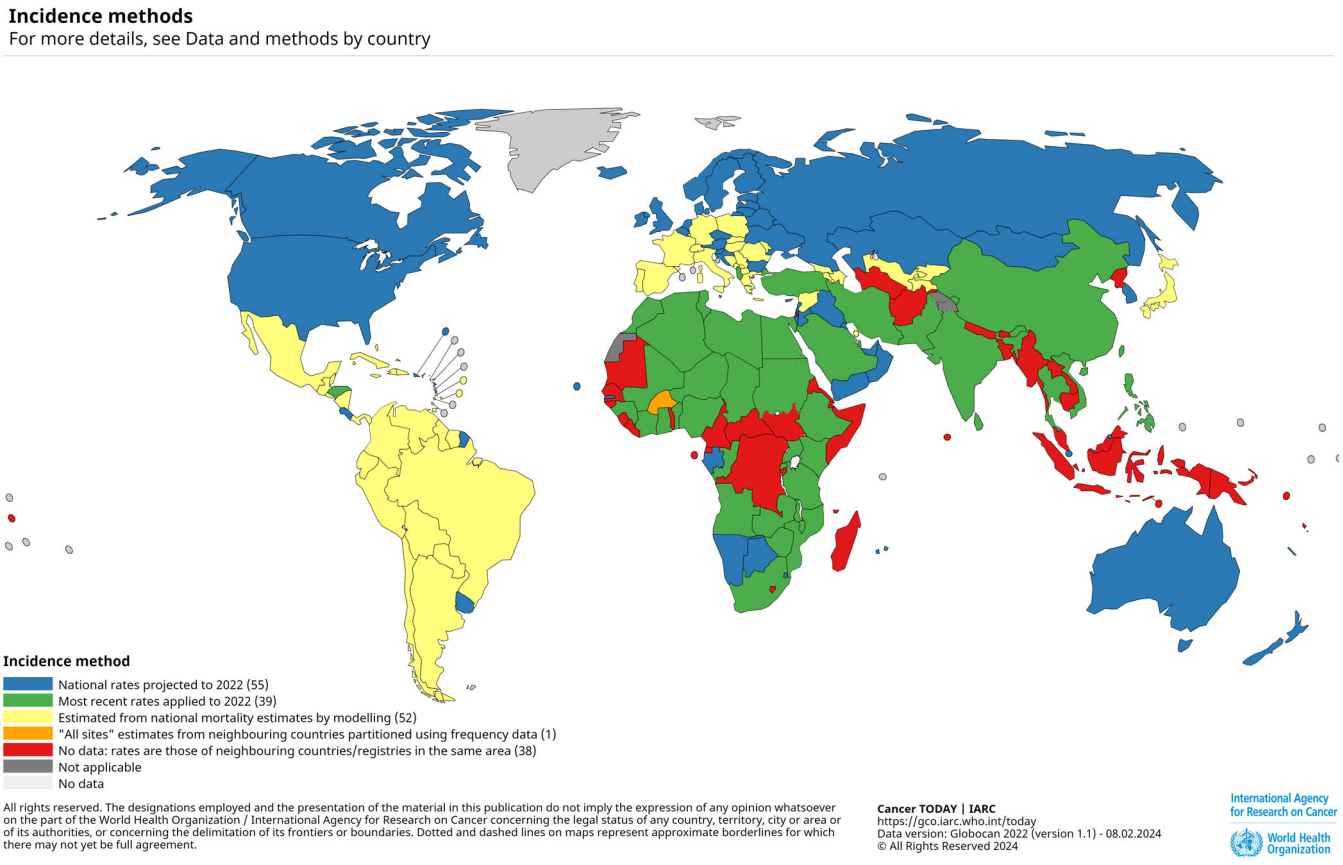

FIGURE 1 Global cancer incidence estimation methods by country for 2022. [Color figure can be viewed at wileyonlinelibrary.com]

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procedures used to estimate incidence and mortality can be found in *Annex C*.

2.3 | Estimates of cancer incidence by countries

The methods employed to estimate the sex- and age-specific incidence rates of cancer within a particular country follow a hierarchical approach according to the available source information. The methods used to estimate the sex- and age-specific incidence rates of cancer in a specific country in 2022 are depicted in Figure 1. Estimates are based on the following broad categories, in order of priority:

1. National rates (or rates based on subnational areas with coverage >50%) projected to 2022 (55 countries).

2a. Subnational rates (from a single registry) applied to the 2022 population (22 countries).

2b. Weighted average of subnational rates (from more than one registry) applied to the 2022 population (17 countries).

3a. National mortality applied to models of incidence:mortality ratios obtained from country-specific cancer registry data (8 countries).

3b. National mortality applied to models of incidence:mortality ratios obtained from cancer registry data in neighbouring countries (9 countries).

3c. National mortality applied to models of incidence:mortality ratios obtained from survival estimates (35 countries).

4. Age- and sex-specific national incidence rates for all cancers combined were obtained by averaging overall rates from neighbouring countries. These rates were then partitioned to obtain the national incidence for specific sites using available cancer-specific relative frequency data in the country (1 country).

5. No data. Rates obtained from neighbouring countries or registries in the same area (38 countries).

2.4 | Estimates of cancer mortality by countries

The methods to estimate cancer-specific mortality depended on the granularity and quality of data available at the national level, as well as the coverage and accuracy of death certificates. Where applicable, the total number of counts were adjusted for under-reporting or incomplete coverage using the percentages supplied by the WHO.[10] Counts classified to "uterus cancer, unspecified" (ICD-10 C55) were reassigned to either "cervix" (C53) or "corpus" (C54) based on the age-specific distribution (*Annexe B*). The hierarchy of approaches utilized to calculate the sex- and age-specific mortality rates of cancer in a particular country are categorized as below and in Figure 2.

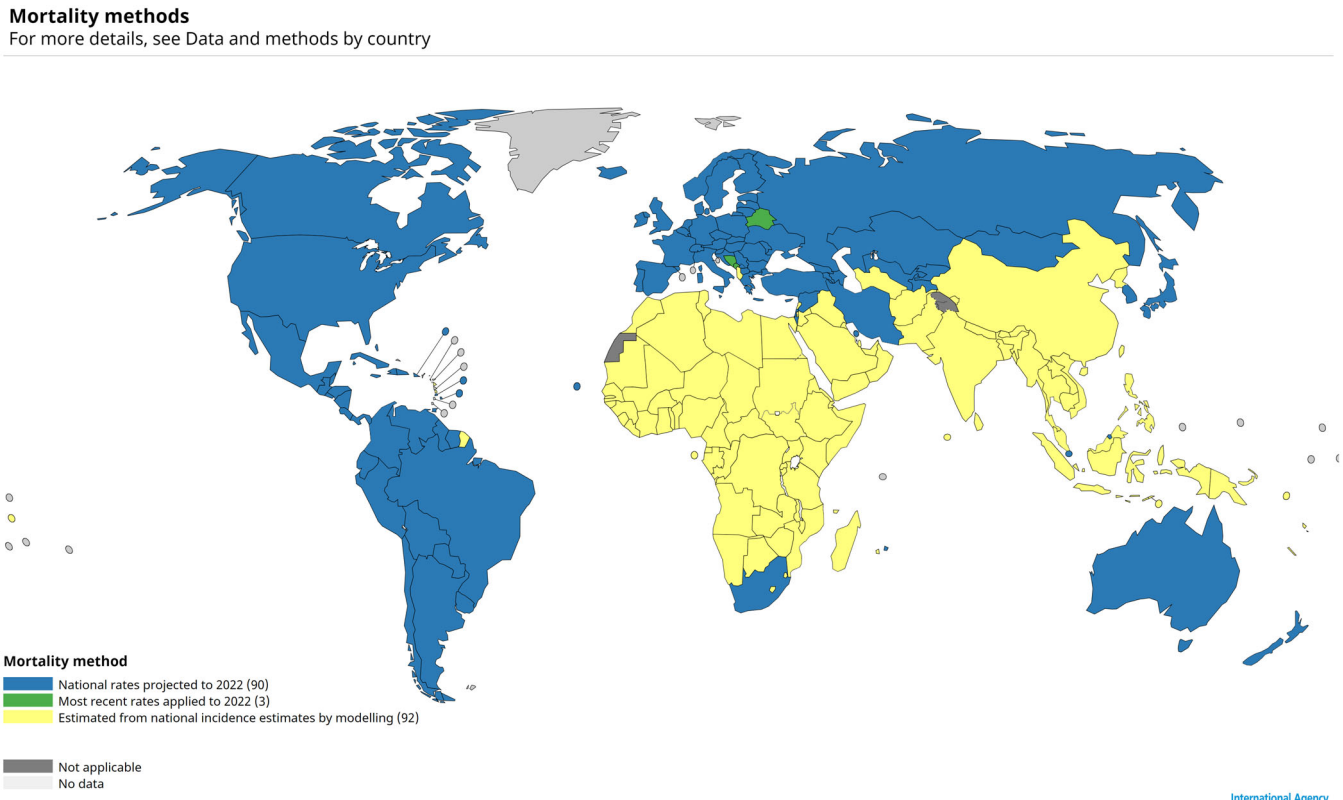

FIGURE 2 Global cancer mortality estimation methods by country for 2022. [Color figure can be viewed at wileyonlinelibrary.com]

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TABLE 1 Estimated new cases and 95% Uncertainty Intervals, age-standardized rates and lifetime cumulative risk by sex and cancer type worldwide, 2022.

	Both sexes				Males				Females			
	ASR	counts	95% UI	cumrisk	ASR	counts	95% UI	cumrisk	ASR	counts	95% UI	cumrisk
Lip, oral cavity	4	389,846	(381,938;397,918)	0.46	5.8	268,999	(266,084;271,946)	0.67	2.29	120,847	(118,767;122,963)	0.26
Salivary glands	0.56	55,083	(52,028;58,318)	0.06	0.66	30,963	(29,806;32,165)	0.07	0.49	24,120	(23,117;25,167)	0.05
Oropharynx	1.1	106,400	(100,655;112,473)	0.13	1.87	86,339	(84,394;88,329)	0.23	0.39	20,061	(19,071;21,103)	0.05
Nasopharynx	1.31	120,434	(115,730;125,330)	0.14	1.92	86,289	(84,478;88,139)	0.21	0.73	34,145	(33,012;35,316)	0.08
Hypopharynx	0.89	86,257	(81,999;90,737)	0.11	1.55	72,077	(70,593;73,592)	0.19	0.29	14,180	(13,540;14,850)	0.03
Oesophagus	4.97	511,054	(502,131;520,136)	0.61	7.58	365,225	(361,794;368,688)	0.93	2.62	145,829	(143,676;148,014)	0.31
Stomach	9.18	968,784	(956,330;981,400)	1.08	12.79	627,458	(622,816;632,135)	1.53	6	341,326	(337,728;344,962)	0.67
Colon	10.69	1,142,286	(1,126,812;1,157,972)	1.22	12.37	609,228	(603,702;614,804)	1.43	9.22	533,058	(527,675;538,496)	1.03
Rectum	7.11	729,833	(718,177;741,678)	0.85	9.06	436,185	(431,844;440,570)	1.10	5.37	293,648	(289,967;297,376)	0.62
Anus	0.54	54,306	(50,781;58,076)	0.06	0.51	24,033	(22,935;25,183)	0.06	0.57	30,273	(28,850;31,767)	0.07
Liver and intrahepatic bile ducts	8.57	866,136	(853,373;879,090)	1.00	12.67	600,676	(595,922;605,468)	1.49	4.76	265,460	(262,152;268,810)	0.55
Gallbladder	1.15	122,491	(118,361;126,765)	0.13	0.88	43,538	(42,365;44,744)	0.10	1.41	78,953	(77,333;80,607)	0.16
Pancreas	4.69	510,992	(500,694;521,501)	0.53	5.45	269,709	(266,040;273,429)	0.64	3.98	241,283	(237,676;244,945)	0.44
Larynx	1.91	189,191	(180,540;198,257)	0.24	3.51	165,794	(163,201;168,428)	0.44	0.45	23,397	(22,388;24,451)	0.05
Trachea, bronchus and lung	23.62	2,480,675	(2,459,654;2,501,876)	2.88	32.07	1,572,045	(1,564,078;1,580,053)	3.88	16.24	908,630	(902,448;914,855)	1.95
Melanoma of skin	3.22	331,722	(318,949;345,006)	0.35	3.66	179,953	(175,182;184,854)	0.40	2.88	151,769	(147,485;156,177)	0.31
Non-melanoma skin cancer	10.37	1,234,533	(1,211,353;1,258,157)	0.98	14.05	744,785	(735,688;753,994)	1.29	7.47	489,748	(482,731;496,867)	0.7
Mesothelioma	0.28	30,633	(27,567;34,040)	0.03	0.42	21,410	(20,109;22,795)	0.05	0.16	9223	(8473;10,040)	0.02
Kaposi sarcoma	0.41	35,813	(31,772;40,368)	0.04	0.56	24,620	(23,014;26,338)	0.05	0.26	11,193	(10,139;12,357)	0.02
Breast	46.82	2,296,840	(2,285,641;2,308,093)	5.06					46.82	2,296,840	(2,285,641;2,308,093)	5.06
Vulva	0.83	47,336	(45,529;49,214)	0.09					0.83	47,336	(45,529;49,214)	0.09
Vagina	0.36	18,819	(17,959;19,720)	0.04					0.36	18,819	(17,959;19,720)	0.04
Cervix uteri	14.12	662,301	(657,146;667,496)	1.50					14.12	662,301	(657,146;667,496)	1.5
Corpus uteri	8.37	420,368	(415,409;425,386)	1.01					8.37	420,368	(415,409;425,386)	1.01
Ovary	6.65	324,603	(320,784;328,467)	0.73					6.65	324,603	(320,784;328,467)	0.73
Penis	0.79	37,700	(36,507;38,932)	0.09	0.79	37,700	(36,507;38,932)	0.09				

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TABLE 1 (Continued)

	Both sexes				Males				Females			
	ASR	counts	95% UI	cumrisk	ASR	counts	95% UI	cumrisk	ASR	counts	95% UI	cumrisk
Prostate	29.42	1,467,854	(1,457,501;1,478,281)	3.68	29.42	1,467,854	(1,457,501;1,478,281)	3.68				
Testis	1.69	72,040	(69,757;74,398)	0.13	1.69	72,040	(69,757;74,398)	0.13				
Kidney	4.42	434,840	(424,177;445,771)	0.51	5.92	277,800	(273,718;281,943)	0.69	3.04	157,040	(153,941;160,202)	0.34
Bladder	5.58	614,298	(599,504;629,457)	0.64	9.31	471,293	(465,875;476,774)	1.05	2.38	143,005	(139,968;146,108)	0.27
Brain, central nervous system	3.47	321,731	(314,207;329,435)	0.35	3.88	173,699	(170,940;176,503)	0.40	3.09	148,032	(145,475;150,634)	0.32
Thyroid	9.12	821,214	(808,835;833,782)	0.91	4.59	206,485	(203,812;209,193)	0.46	13.64	614,729	(609,950;619,546)	1.35
Hodgkin lymphoma	0.95	82,469	(78,025;87,167)	0.08	1.14	48,774	(47,136;50,469)	0.10	0.77	33,695	(32,257;35,197)	0.07
Non-Hodgkin lymphoma	5.57	553,389	(542,283;564,722)	0.60	6.64	311,375	(307,304;315,499)	0.72	4.58	242,014	(238,310;245,775)	0.49
Multiple myeloma	1.79	187,952	(181,507;194,626)	0.22	2.12	103,805	(101,448;106,217)	0.25	1.51	84,147	(81,966;86,386)	0.18
Leukaemia	5.26	487,294	(477,269;497,529)	0.50	6.23	278,120	(274,403;281,887)	0.59	4.38	209,174	(205,885;212,515)	0.41
Other specified cancers	7.29	704,447	(693,007;716,076)	0.76	8.45	390,243	(386,077;394,454)	0.88	6.26	314,204	(310,343;318,113)	0.64
Unspecified sites	4.48	454,535	(445,854;463,385)	0.49	5.1	243,096	(239,991;246,242)	0.56	3.94	211,439	(208,422;214,500)	0.42
All cancers	196.88	19,976,499	(19,914,145;20,039,048)	20.03	212.64	10,311,610	(10,289,327;10,333,941)	21.80	186.33	9,664,889	(9,643,101;9,686,727)	18.52
All cancers excl. non-melanoma skin cancer	186.51	18,741,966	(18,682,643;18,801,478)	19.23	198.59	9,566,825	(9,545,824;9,587,872)	20.78	178.86	9,175,141	(9,154,199;9,196,130)	17.94

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TABLE 2 Estimated deaths and 95% Uncertainty Intervals, age-standardized rates, and lifetime cumulative risk by sex and cancer type worldwide, 2022.

	Both sexes				Males				Females			
	ASR	counts	95% UI	cumrisk	ASR	counts	95% UI	cumrisk	ASR	counts	95% UI	cumrisk
Lip, oral cavity	1.92	188,438	(182,938;194,104)	0.22	2.81	130,808	(128,760;132,888)	0.32	1.07	57,630	(56,204.7;59,091.4)	0.12
Salivary glands	0.23	23,942	(21,900;26,174)	0.03	0.29	13,989	(13,193;14,833)	0.03	0.18	9953	(9306.5;10,644.4)	0.02
Oropharynx	0.53	52,305	(48,324;56,613)	0.06	0.91	42,818	(41,468;44,211)	0.11	0.18	9487	(8825.5;10,198.1)	0.02
Nasopharynx	0.77	73,482	(69,495;77,698)	0.09	1.18	54,104	(52,576;55,677)	0.14	0.39	19,378	(18,473.3;20,327)	0.04
Hypopharynx	0.41	40,902	(37,647;44,438)	0.05	0.73	34,564	(33,437;35,729)	0.09	0.12	6338	(5874.8;6837.7)	0.01
Oesophagus	4.26	445,391	(436,554;454,407)	0.50	6.54	318,433	(315,018;321,885)	0.78	2.22	126,958	(124,833.7;129,118.4)	0.25
Stomach	6.09	660,175	(649,189;671,347)	0.69	8.57	427,575	(423,484;431,705)	0.98	3.93	232,600	(229,425.9;235,818)	0.42
Colon	4.71	538,167	(526,889;549,687)	0.47	5.55	283,784	(279,746;287,881)	0.57	3.99	254,383	(250,448.5;258,379.3)	0.39
Rectum	3.14	343,817	(335,478;352,363)	0.34	4.09	205,117	(202,004;208,278)	0.45	2.31	138,700	(136,063.1;141,388)	0.25
Anus	0.21	22,035	(19,929;24,363)	0.02	0.22	10,874	(10,158;11,640)	0.02	0.2	11,161	(10,366.8;12,016.1)	0.02
Liver and intrahepatic bile ducts	7.37	758,725	(746,051;771,615)	0.85	10.9	521,826	(517,095;526,601)	1.26	4.13	236,899	(233,569.3;240,276.1)	0.47
Gallbladder	0.83	89,055	(85,385;92,882)	0.09	0.63	31,406	(30,363;32,484)	0.07	1.02	57,649	(56,216.7;59,117.8)	0.11
Pancreas	4.21	467,409	(456,923;478,136)	0.47	4.95	247,589	(243,840;251,396)	0.57	3.53	219,820	(216,160.7;223,541.2)	0.38
Larynx	1.01	103,359	(97,090;110,033)	0.12	1.88	90,384	(88,406;92,406)	0.23	0.23	12,975	(12,238.6;13,755.8)	0.03
Trachea, bronchus and lung	16.76	1,817,469	(1,797,364;1,837,799)	1.97	24.77	1,233,241	(1,225,739;1,240,789)	2.91	9.79	584,228	(578,825.1;589,681.3)	1.11
Melanoma of skin	0.53	58,667	(54,151;63,559)	0.05	0.65	33,160	(31,411;35,006)	0.07	0.43	25,507	(24,046;27,056.8)	0.04
Non-melanoma skin cancer	0.59	69,416	(65,482;73,587)	0.05	0.77	39,688	(38,203;41,231)	0.07	0.45	29,728	(28,444.5;31,069.4)	0.04
Mesothelioma	0.22	25,371	(22,376;28,767)	0.02	0.35	18,082	(16,813;19,447)	0.03	0.13	7289	(6580;8074.3)	0.01
Kaposi sarcoma	0.19	16,169	(13,639;19,168)	0.02	0.25	10,629	(9600;11,768)	0.02	0.13	5540	(4834.5;6348.5)	0.01
Breast	12.65	666,103	(660,000;672,262)	1.36					12.65	666,103	(660,000.1;672,262.4)	1.36
Vulva	0.3	18,579	(17,438;19,794)	0.03					0.3	18,579	(17,438.5;19,794.1)	0.03
Vagina	0.15	8240	(7658;8866)	0.02					0.15	8240	(7658.2;8866)	0.02
Cervix uteri	7.08	348,874	(344,885;352,909)	0.79					7.08	348,874	(344,885.2;352,908.9)	0.79
Corpus uteri	1.72	97,723	(95,204;100,309)	0.20					1.72	97,723	(95,203.6;100,309.1)	0.2
Ovary	3.97	206,956	(203,693;210,271)	0.46					3.97	206,956	(203,693.2;210,271)	0.46
Penis	0.28	13,738	(13,012;14,505)	0.03	0.28	13,738	(13,012;14,505)	0.03				
Prostate	7.27	397,430	(391,946;402,991)	0.61	7.27	397,430	(391,946;402,991)	0.61				
Testis	0.21	9068	(8348;9850)	0.02	0.21	9068	(8348;9850)	0.02				
Kidney	1.46	155,953	(149,376;162,819)	0.16	2.04	100,343	(97,828;102,923)	0.22	0.95	55,610	(53,709;57,578)	0.1
Bladder	1.82	220,596	(211,981;229,561)	0.17	3.11	165,672	(162,496;168,910)	0.28	0.8	54,924	(53,047;56,867)	0.07
Brain, central nervous system	2.59	248,500	(241,347;255,864)	0.28	3.05	139,823	(137,190;142,506)	0.33	2.16	108,677	(106,295;111,112)	0.23

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TABLE 2 (Continued)

	Both sexes				Males				Females			
	ASR	counts	95% UI	cumrisk	ASR	counts	95% UI	cumrisk	ASR	counts	95% UI	cumrisk
Thyroid	0.44	47,507	(44,496;50,722)	0.05	0.35	17,241	(16,379;18,148)	0.04	0.53	30,266	(29,057;31,525)	0.06
Hodgkin lymphoma	0.24	22,733	(20,519;25,186)	0.02	0.31	13,674	(12,840;14,562)	0.03	0.18	9059	(8356;9822)	0.02
Non-Hodgkin lymphoma	2.38	250,679	(243,092;258,503)	0.24	2.95	143,740	(140,951;146,584)	0.30	1.88	106,939	(104,438;109,499)	0.19
Multiple myeloma	1.11	121,388	(116,118;126,898)	0.13	1.33	66,966	(65,042;68,947)	0.15	0.92	54,422	(52,631;56,274)	0.1
Leukaemia	3.09	305,405	(297,273;313,759)	0.29	3.7	173,289	(170,286;176,345)	0.35	2.55	132,116	(129,429;134,859)	0.24
Other specified cancers	3.96	401,598	(392,776;410,618)	0.41	4.68	222,441	(219,239;225,690)	0.48	3.32	179,157	(176,169;182,195)	0.33
Unspecified sites	3.83	408,468	(398,478;418,708)	0.41	4.46	218,788	(215,237;222,398)	0.48	3.29	189,680	(186,190;193,236)	0.34
All cancers	91.71	9,743,832	(9,699,262;9,788,607)	9.61	109.75	5,430,284	(5,414,116;5,446,500)	11.40	76.85	4,313,548	(4,298,560;4,328,588)	7.98
All cancers excl. non-melanoma skin cancer	91.12	9,674,416	(9,630,027;9,719,010)	9.57	108.98	5,390,596	(5,374,497;5,406,743)	11.33	76.41	4,283,820	(4,268,889;4,298,803)	7.94

1. National rates projected to 2022 (90 countries).
2a. National rates applied to 2022 (3 countries).
3b. National incidence estimates applied to models of mortality: incidence ratios obtained from cancer registry data in neighbouring countries (1 country).
3c. National incidence estimates applied to models of mortality: incidence ratios obtained from survival estimates (91 countries).

We used weighted regression with the loess function to minimize random fluctuations in the estimated age-specific rates. For the 20 world regions, estimates were derived as a population-weighted average of the incidence and mortality rates of the constituent countries. These rates were subsequently applied to the respective populations to estimate the number of new cases and deaths in 2022. We calculated age-standardized rates (ASR per 100,000 person-years) using the direct standardization method (World Standard Population) initially proposed by Segi[11] and later modified by Doll.[12] Additionally, to enhance comparability, we estimated the cumulative risk of developing or dying from cancer before the age of 75 years, assuming an absence of competing causes of death. These estimates were computed using the age-specific rates and are presented as percentages.

2.5 | Uncertainty intervals

95% uncertainty intervals accompany the estimated counts of new cases and deaths by country, cancer site and sex, utilizing the standard error (SE) of the crude incidence and mortality rates. To account for uncertainty in the source information, the standard error is adjusted for the: (a) degree of national coverage; (b) timeliness, and (c) data quality considerations including completeness and accuracy. To produce the standard error, penalties were used for each of the indicators above. Further information on computation of the standard error can be found in *Annex D*.

3 | RESULTS

Tables 1 and 2 present the estimated new cases of cancer and cancer deaths and the corresponding age-standardized rates (ASR) across cancer types in 2022, along with the associated 95% uncertainty intervals (UI), in both sexes, and for males and females. There were 18.74 million (95%UI: 18.68–18.80 million) cancer diagnoses worldwide excluding non-melanoma skin cancer in 2022 and 9.74 million (9.70–9.79 million) cancer deaths (9.67 million excluding non-melanoma skin cancer). In males, lung cancer was the most frequently diagnosed cancer worldwide (1.57 million [1.56–1.58]), followed by prostate (1.47 million [1.46 – 1.48]), and stomach (0.63 million [0.62 – 0.63]). The most common causes of male cancer death were from lung (1.23 million [1.22 – 1.24]), liver (0.52 million [0.52 – 0.53]), and stomach (0.43 million [0.42 – 0.43]). With 2.30 million (2.28–2.31) cases diagnosed in females, breast cancer was the most common cancer, followed by lung (0.91 million [0.90–0.91]) and cervix uteri (0.66 million [0.66 – 0.67]).

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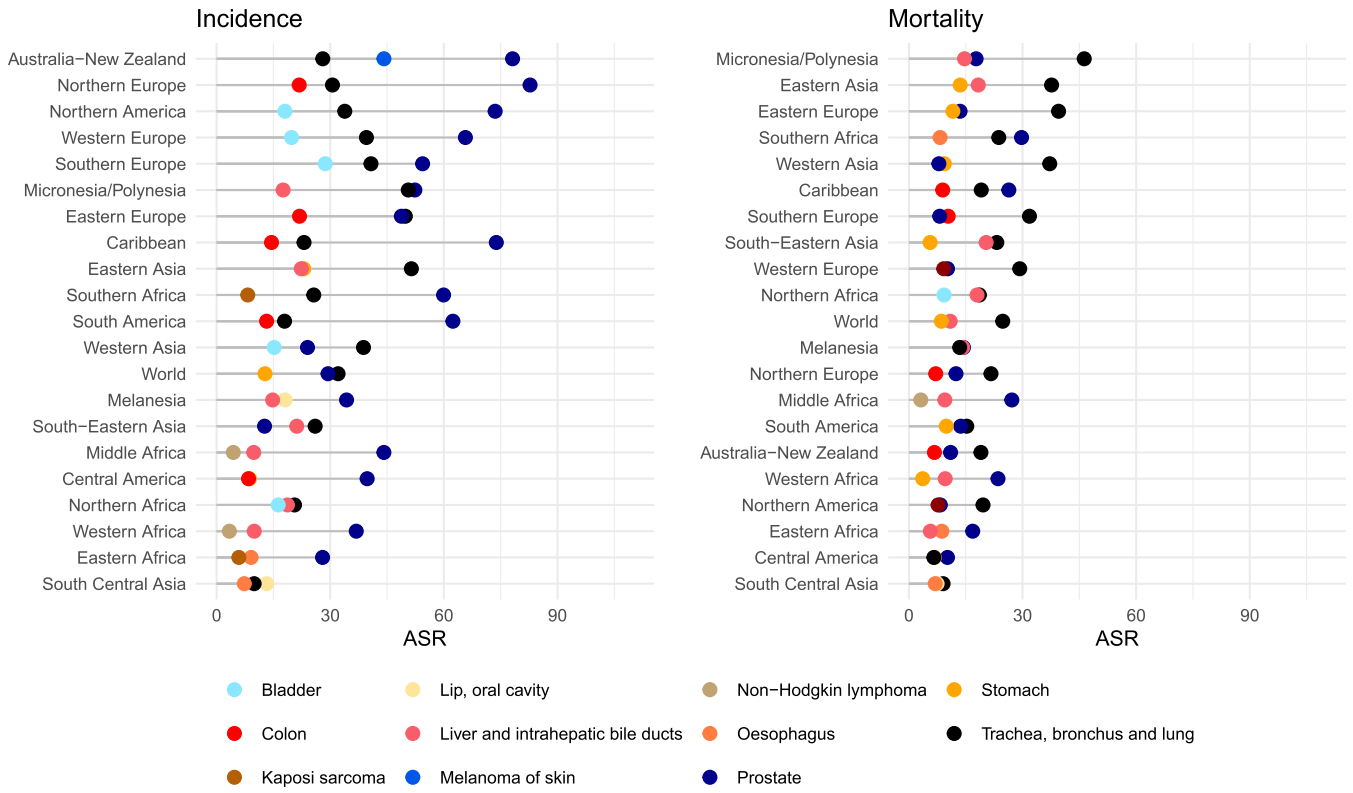

FIGURE 3 Leading types of cancer in terms of new cases (incidence) and deaths (mortality) in males in each of the 20 world regions in 2022. [Color figure can be viewed at wileyonlinelibrary.com]

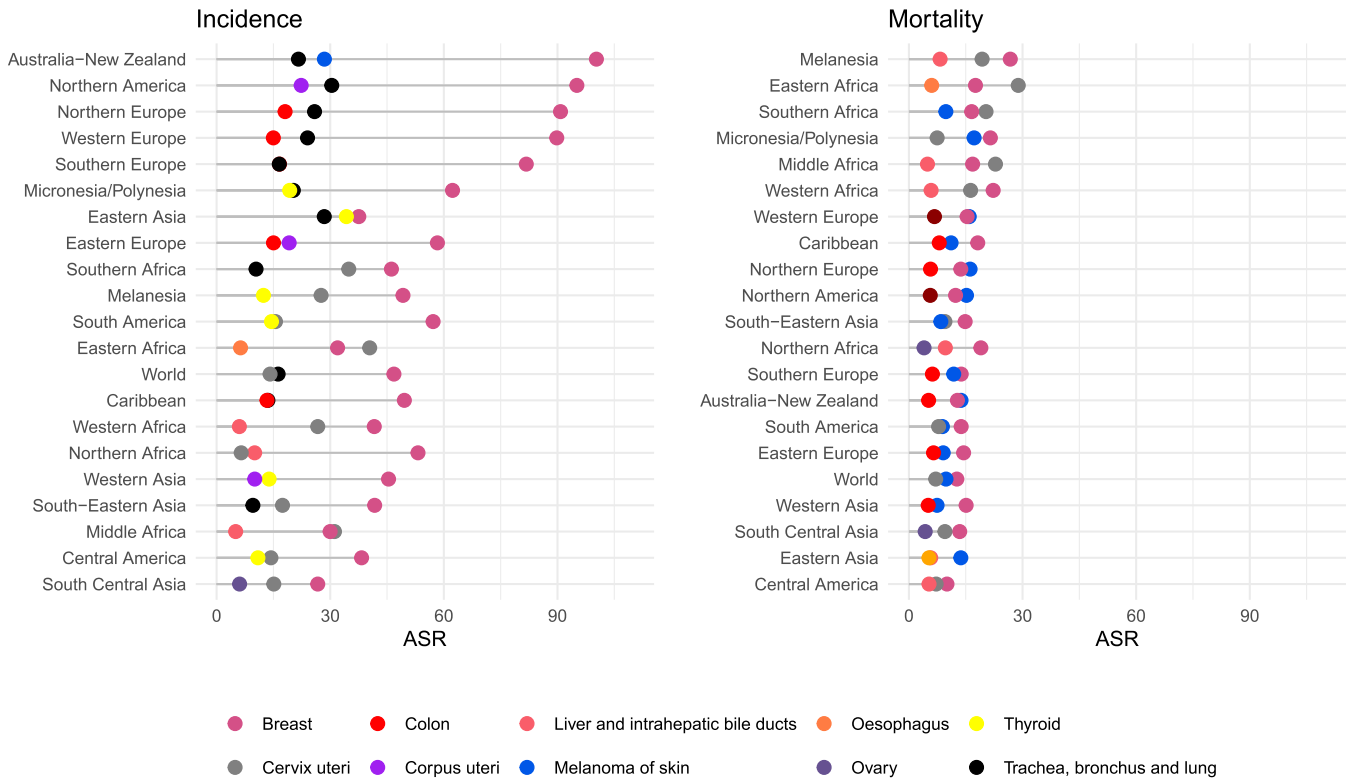

FIGURE 4 Leading types of cancer in terms of new cases (incidence) and deaths (mortality) in females in each of the 20 world regions in 2022. [Color figure can be viewed at wileyonlinelibrary.com]

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Female breast cancer remains the leading cause of cancer death in females worldwide (0.67 million [0.66 – 0.67]), followed by lung (0.58 million [0.58 – 0.59]) and cervix uteri (0.35 million [0.34 – 0.35]).

Figures 3 and 4 highlight the disparity between cancer incidence and mortality across different world regions in 2022, by sex. In males, prostate cancer remains the most common cancer site in 14 world regions, followed by lung cancer (six regions). The most frequent cause of cancer death was lung cancer (13 regions), followed by prostate cancer (six regions). Among females, breast cancer was the most common cancer in 17 regions. In five African regions and in two Asian regions, cervical cancer was either the most common, or second most common cancer. Breast cancer was the most common cause of death in 14 world regions in 2022, followed by cervical cancer in three further regions.

Annex E, Figure S1A–C shows the estimated counts of new diagnoses and deaths worldwide in 2022. Almost half of all cancers diagnosed worldwide occurred in Asia (49.2%), particularly in Eastern Asia, which includes China, followed by Europe (22.4%) and the Americas (21.1%). In terms of mortality, of the 9.7 million cancer deaths, almost three-fifths occurred in Asia (56.1%), followed by Europe (20.4%) and the Americas (14.9%). Annex E, Figure S2A–C shows the percentages of new cases and deaths by cancer type by sex. Among males, cancers of the lung, prostate, and colorectum accounted for the highest proportions of disease: 15.3%, 14.2%, and 10.4%, respectively. In terms of mortality, lung (22.7%), liver (9.6%), and colorectum (9.2%) were the leading causes of cancer death. Among females, breast (23.8%) and lung (9.4%) were the most frequent cancers in 2022, with breast (15.4%), lung cancer (13.6%), and colorectum (9.4%) being the most common causes of cancer death.

4 | DISCUSSION

This article documents the data sources and methods used by IARC to compile national cancer incidence, and mortality estimates in 185 countries in 2022. The resulting GLOBOCAN database is a snapshot of the cancer burden worldwide, available online in visual and tabular formats on the GCO (https://gco.iarc.who.int/today/en). While the estimation methods have been refined over the past decades to accommodate the increasing availability and quality of data, the core methodological principles remain unchanged; where feasible, national incidence estimates are derived from local sources of cancer incidence, namely PBCR (national or sub-national), while corresponding mortality estimates are based primarily on national vital registration data. The methodology seeks to be both reproducible and transparent at the country level, in alignment with the FAIR principles in epidemiology.[13] The European estimates were developed in collaboration with the Joint Research Centre (JRC) of the European Commission (as they were for 2018 and 2020), and also disseminated on the European Cancer Information System platform (ECIS, https://ecis.jrc.ec.europa.eu/).[14] At a country level, the Chinese estimates were developed in collaboration with the National Central Cancer Registry programme that continues to expand cancer registration towards national coverage,[15] while the estimates in South Africa were undertaken with the National Cancer Registry (https://www.nicd.ac.za/centres/national-cancer-registry/).

The accuracy of IARC's cancer estimates depend on several factors outside the scope of statistical methodologies. Given the varying coverage, timeliness and quality of the recorded incidence and mortality data available in a given country, the estimation of uncertainty intervals (UI) aims to capture additional variation linked to the robustness of the source information. Penalties are applied to adjust the standard error for each indicator over and above the inherent random variation, with the UI widening when recorded incidence or mortality data covered lower proportions of the total population, lacked timeliness, or were considered of poorer quality.

Screening for asymptomatic disease, along with the detection of incidental (asymptomatic) disease due to the use of high-resolution imaging techniques may account for some of the variation in incidence rates.[16] The estimation of new cases of non-melanoma skin cancer poses specific difficulties given variations in diagnostic and registration practice in different countries and world regions, leading to variations in the completeness and accuracy of the incidence burden of non-melanoma skin cancer as recorded by PBCR. Our estimates are based on cancer registry information on the first occurrence of the diagnosis, excluding basal-cell carcinoma.[17]

Considerable caution is advised when comparing 2022 estimates to previous sets of estimates. Changes in incidence and mortality counts or rates are partly attributable to the improved availability and quality of incidence and mortality data worldwide. In the absence of national or subnational data that can be projected to 2022, site-, sex, and age-specific mortality-to-incidence (M:I) ratios from PBCR were applied to national mortality data to estimate national incidence; this method has been validated and applied extensively over several decades, and potential sources of bias have been detailed elsewhere.[5–7] Future activities will include a focus on strengthening the collection of accurate follow-up of cancer patients to obtain robust survival estimates that can be used to derive mortality.[7]

The COVID-19 pandemic disrupted cancer registry operations and notable delays in cancer diagnosis and screening have been reported in the US,[18] with data from Surveillance, Epidemiology, and End Results (SEER) indicating a shortfall in cancer incidence of around 10% in 2020.[19] Other countries, such as Australia, Cuba, and the Netherlands also reported a drop in cancer incidence ranging from circa 10% to 45% in 2020 compared with previous years.[20–22] We did not account for the effects of pandemic when compiling the 2022 estimates, which are based on incidence and mortality data up until 2019. While this may have led to overestimation of rates in some countries, studies have also shown that for many countries the reduction in cancer incidence appears a short-term effect linked to the non-pharmaceutical interventions implemented during the height of the pandemic.[23]

IARC's GLOBOCAN estimates are a snapshot of the cancer incidence and mortality in a single year and seek to support prioritization efforts in cancer control and cancer prevention worldwide. It is important to note such national estimates for a given year are not intended

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and cannot replace the continuous data collection undertaken by PBCR and national vital registration systems, which are essential for the local monitoring and evaluation of cancer control plans. Yet, the rising demand for PBCR data contrasts with the operational challenges faced by many registries, including insufficient and short-term funding, and a shortage of trained personnel. With only one in three countries currently able to report high-quality data, the IARC-WHO's Global Initiative for Cancer Registry Development (GICR, https://gicr.iarc.who.int) is an essential partnership aiming to drive down inequities in the availability of cancer incidence and survival data and ensure there are measurable improvements in the quality and use of such data.

AUTHOR CONTRIBUTIONS

Adalberto M. Filho: Writing – original draft; visualization; writing – review and editing; supervision; methodology; investigation; formal analysis; conceptualization. **Mathieu Laversanne:** Methodology; validation; visualization; software; formal analysis; writing – review and editing. **Jacques Ferlay:** Conceptualization; methodology; visualization; validation; investigation. **Murielle Colombet:** Methodology; validation; data curation; software; writing – review and editing. **Marion Piñeros:** Conceptualization; investigation; writing – review and editing; validation. **Ariana Znaor:** Conceptualization; investigation; writing – review and editing; validation. **Donald M. Parkin:** Methodology; funding acquisition; writing – review and editing; conceptualization. **Isabelle Soerjomataram:** Conceptualization; investigation; methodology; validation; writing – review and editing. **Freddie Bray:** Conceptualization; investigation; funding acquisition; methodology; validation; visualization; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

These datasets are either in the public domain or available upon request to their owners. The full results of the study are available at the Global Cancer Observatory (GCO) (https://gco.iarc.fr). Further information is available from the corresponding author upon reasonable request.

DISCLAIMER

Where authors are identified as personnel of the International Agency for Research on Cancer/World Health Organization, the authors alone are responsible for the views expressed in this article and they do not necessarily represent the decisions, policy or views of the International Agency for Research on Cancer/World Health Organization.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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